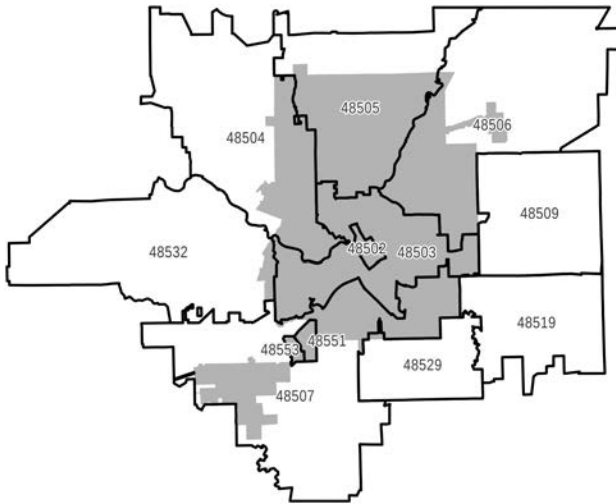




**Figure 5.4.** The misalignment of the municipal boundary of the city of Flint, Michigan (grey polygon), with the ZIP code boundaries used for collation of health data (black boundaries and numeric labels).

difficult to obtain or use more precise geospatial information, it seems that information as badly flawed as it was in this case (see also Grubestic & Matisziw, 2006) was no better than no information at all might have been.

## THE ARBITRARINESS OF BOUNDARIES

As with scale (see Chapter 3), the social contestation and production of boundaries and territories is not explicitly represented in any concrete way in geographical computation, but defined boundaries and the resulting spatial units can have profound effects on analytical outcomes.

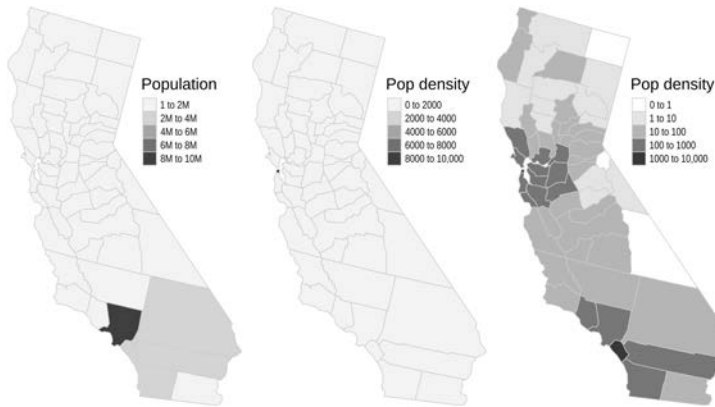
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visualize data and perform spatial data analysis on data with associated ZIP codes (see Krieger et al., 2002).

Choosing which areas to use in a particular context, or even designing boundaries appropriate to a particular project, is a frequently encountered challenge in giscience. Ideally, the bounded areas used in any analysis would be real—bona fide boundaries to use Smith’s term—but as we have seen, there are often really no such boundaries available, unless we are exploring specific technical questions about particular bureaucracies. So, it might make sense to work with ZIP codes if the questions at hand are about the speed of mail delivery, or with school enrollment zones if the questions are about outcomes in schools. More general questions about social, political, economic, and cultural geographies may not have any associated geographies that are fit for analytical purpose.

Furthermore, there is no single list of desirable characteristics of a set of geographies that can guarantee their usefulness for a wide range of uses, although we *can* identify characteristics that are generally *not* desirable! All else equal, it is better if areas have roughly similar populations, or more generally, populations that are broadly comparable to one another. In this context, the widespread use of counties and states for mapping geographical patterns in the United States provides a good example of *very bad* spatial units—at least from a technical perspective. The maps in Figure 5.5, based on counties in California, give some sense of the extent of the problem. Counties are just not a comparable set of things—their wildly divergent populations are almost their defining feature.

An associated problem this causes is that counties with smaller populations tend to dominate the extreme positions when we measure the rate of occurrence of anything—such as disease incidence, voter turnout, unemployment, and so on. This is because large populations will tend toward the mean rate of occurrence. These problems are particularly marked in dealing with rare events in spatial epidemiology (see chapter 5 in Cromley & McLafferty, 2012). Disentangling such artifacts of the spatial units from real effects associated with differences between rural and urban places can be challenging. Another difficulty with highly variable spatial units is that large-area, sparsely populated polygons dominate conventional maps, while densely populated areas disappear from view (see the middle panel of Figure 5.5). Cartograms offer one possible way to



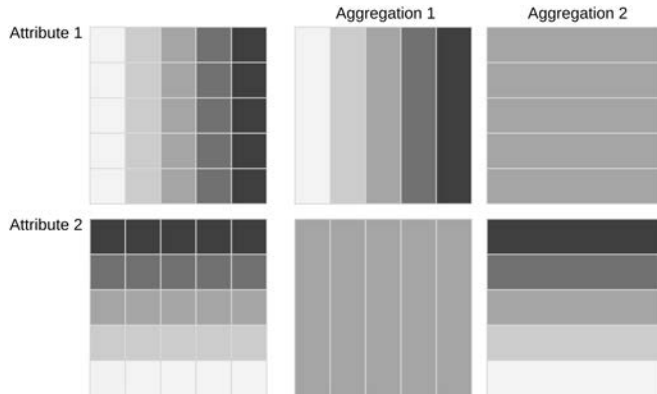
**Figure 5.5.** The challenge of maps when areas have very different base populations. The left-hand map shows the problem: Populations of the counties of California range from around 8,000 to almost 10,000,000. Mapping population density does not help much (middle map). Changing to a logarithmic scale (right-hand map) yields a more useful map, but also emphasizes just how different the counties are from one another.

mitigate this problem (see Figure 2.2) at the expense of less immediately accessible maps and visualizations.

### The Modifiable Areal Unit Problem

These challenges point to the need for careful consideration of the system of geographies to be used in particular studies, which leads directly to consideration of the *modifiable areal unit problem* (MAUP). It is obvious that any system of geographies (usually polygons) applied to a region and used to aggregate statistical or count data translates a set of underlying observations into a table of numbers.<sup>15</sup> Less obvious (perhaps) is that the particular set of polygons used can alter the resulting distributions and patterns. A simple example (see Figure 5.6) suffices to demonstrate

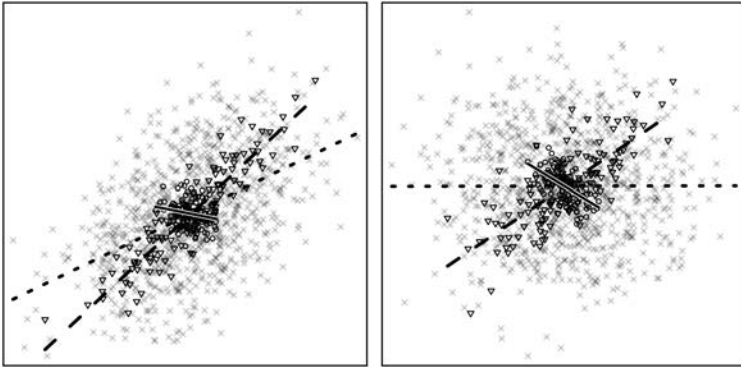
<sup>15</sup>Another instance of the linking of “territory to other things” of Wood (2010b) in practice.



**Figure 5.6.** Simple illustration of the modifiable areal unit problem. Two attributes (the column and row numbers of the squares) are averaged across columns and rows respectively. In each aggregation the pattern of only one of the two attributes is preserved.

the effect. This is an extreme case to demonstrate the problem, but the effect is real, and in practice much more subtle than this. Perhaps surprisingly, the effect impacts not only simple summary values of attributes, but also secondary measures like the correlations between variables. This was dramatically illustrated by Openshaw and Taylor (1979) who showed how different aggregations of smaller polygons into larger ones could lead to the apparent correlation between two variables ranging anywhere between  $-1$  and  $+1$ .

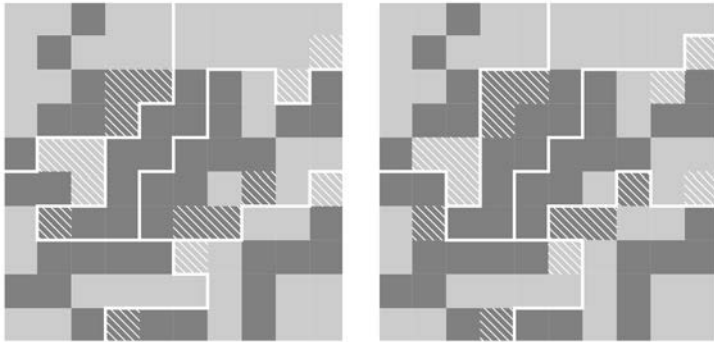
How this can happen is illustrated in Figure 5.7. Depending on whether observations that are similar or dissimilar are combined, an initial correlation can be strengthened or even changed in direction, and it is also possible for data that are not correlated at all to appear correlated after aggregation. The example in the figure is for non-spatial data. In a geographical setting, the impact of aggregating data from neighboring zones into larger agglomerations depends on the scales at which similarities and differences manifest in the data, that is on the



**Figure 5.7.** How aggregating data can increase or decrease correlation. Original data are shown in the background as crosses with the associated best-fit line dotted. In the left-hand panel the original correlation for these data is 0.47. Aggregating groups of eight similar observations yields the data shown as triangles, which have a correlation of 0.92 (the dashed line), while aggregating sets of eight dissimilar observations gives the data shown as circles with a *negative* correlation -0.21 (the solid line). In the right-hand panel initially uncorrelated data aggregate to correlations of 0.77 and -0.55, respectively!

scale, extent, and sign of any spatial autocorrelation. This aggregation behavior in data is an instance of an ecological correlation (Robinson, 1950) and the effect was familiar long before Openshaw and Taylor's experiments (Gehlke & Biehl, 1934). However, often finer-grained data are unavailable, and it may thus be impossible to know the degree to which observed correlations relate to effects the correlation statistics are intended to estimate.

The MAUP is driven by two different effects. First is an aggregation effect, which is the ecological correlation already considered, and a clear example of a scale effect (see §Scale-dependencies, Chapter 3). Second is a zoning effect, which reflects differences that can arise aggregating data at a single level, but in different ways, by drawing different lines to delineate different sets of polygons. It is the zoning effect that was explored by Openshaw and Taylor (1979) and that is illustrated in Figure 5.6. The



**Figure 5.8.** The zoning effect as a gerrymandering effect. In the left-hand panel the light gray group is in a narrow majority in four of five districts, while in the right-hand panel the result is reversed, after only a few sub-areas (hatched white) are swapped between districts.

zoning effect could equally be called the “gerrymandering effect” as it centers on how aggregate outcomes change as boundaries between zones shift (see Figure 5.8).

*Gerrymandering* is the political process by which electoral district boundaries are manipulated to make election outcomes less uncertain for the parties involved in the design of the districts. Given accurate information about the voting preferences of populations, an effectively gerrymandered map of electoral districts is one that makes the most efficient use of the votes available to the party designing the map. This involves some combination of “packing” an opponent’s votes into large safe majorities where many of their votes are wasted, because they are not required to guarantee a win, or by “cracking” concentrations of the opponent’s voter base by splitting them across several electoral districts (see Monmonier, 2001, especially pp. 8–12). Gerrymandering can be significantly refined using GIS software.

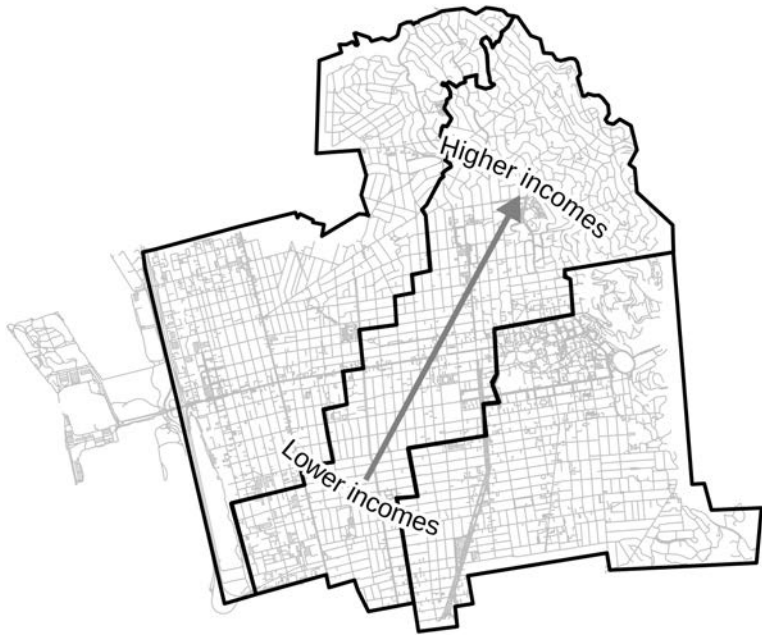
Concerns about gerrymandering, especially in the United States, have led to numerous ideas for assessing how fair a given map is. Some approaches center on assessing the efficiency of votes for different parties,

that is, how many votes each party had to accrue per elected representative. If one party has to stack up more votes than the other to win seats, then perhaps it points to deliberately biased design of districts. Other approaches focus on the *shape* of districts, suggesting that extremely convoluted district boundaries are an indicator of deliberate, and by implication, bad intention in the design of districts, and that fair districts would have less convoluted, substantially convex shapes. Many of these approaches seem naïve at best. If the desired outcome is that the numbers of elected representatives be proportional to the votes cast for each party, then proportional voting systems are guaranteed to achieve that outcome. What the MAUP tells us is that outcomes in first-past-the-post representative systems inevitably depend on where, by whom, and how the electoral lines are drawn. Meanwhile, it is unclear what would constitute fair design of electoral districts, a question unavoidably entangled with the question of what constitutes a community of interest, which is the legal notion (in the United States) that has become relevant to these questions (Morrill, 1987; Forest, 2004).

Given the strong negative connotations of the term gerrymander, it is worth noting that the design of zoning systems can also aim to achieve equitable outcomes. An example, current in the 2010s, is the middle school zones of Berkeley, California, shown in Figure 5.9. This is related to a suggestion made by Openshaw

that the MAUP is not so much an insoluble problem but rather a powerful analytical tool ideally suited for probing the structure of areal data sets. The growing speed of computers opens up the tremendous potential offered by heuristic solution procedures, such as the AZP [automating zoning procedure], to identify the most appropriate zoning systems for any particular purpose (1983, p. 38).

In other words, instead of treating the MAUP as an inconvenience, consider it an opportunity to get a better understanding of the geography of a study area by partitioning it in ways appropriate to the topic at hand. Openshaw (1983) also described an outline AZP drawing on earlier work (Openshaw, 1977), and some of these ideas were taken up in designing flexible output geographies for census data (Openshaw & Rao, 1995;



**Figure 5.9.** The (approximate) middle school enrollment zones in Berkeley, California, c. 2015. Each zone spans neighborhoods across levels of the socio-economic hierarchy in the city. They are an attempt to mix school populations on that basis.

Martin, 1998). More recently, perhaps following the development of efficient algorithms for partitioning networks (see §[Connection, Disconnection, and Communities](#), Chapter 6), these ideas have resurfaced (see, e.g., Poorthuis, 2018).

### Regionalizing Space

The inversion of the MAUP into a problem of zone design leads directly to the question of how we can partition a region into meaningful